Chartered Institute of Technology

Model Paper (B.Tech- IInd Sem)

Unit -1

Part -A(Very Short Ans Type Question)

1. Define rank of a matrix.

2. Define consistent and inconsistent of liner equations.

3. State C-H Theorem.

4. Find the Eigen value of matrix A= $\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

5. Define consistent and inconsistent of a linear equation.

Part -B(Short Ans Type Question)

1. Find the rank of following matrics

(i) A= $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ (ii) A= $\begin{bmatrix} 2 & 3 & 4 & -1 \\ 5 & 2 & 0 & -1 \\ -4 & 5 & 12 & -1 \end{bmatrix}$

2. Reduced to a Normal form of a matrix A and find its rank also, if

A= $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$

3. Check the consistency of the following linear equations and the solution of x y, and z.

(i) $x+y+z=3$, $x+2y+3z=4$, $x+4y+9z=6$

(ii) $3x+3y+2z=1$, $x+2y=4$, $10y+3z=-2$, $2x-3y-z=5$

4, Find the Eigen value and Eigen vector of (i) A= $\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & -6 \\ 2 & -2 & 3 \end{bmatrix}$ (ii) A= $\begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$

5. Using Cayley Hamilton theorem, find A^4 If A= $\begin{bmatrix} 2 & 1 \\ 5 & -2 \end{bmatrix}$

Part -C (Long Ans Type Question)

1. For a matrix A = $\begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 1 & 4 \end{bmatrix}$, Find non singular matrices P and Q such that PAQ is in normal form, Find the rank also.

2. For what value of parameter λ and μ do the system of equation have (i) unique solution (ii) No solution (iii) more than one solution.

$$x+y+z=6 \quad , \quad x+2y+6z=10 \ , \ x+2y+\lambda z=\mu$$

3. Show that the system of equations can possess a non trivial solution only if $\lambda=1$ and $\lambda=-3$. obtained the general solution of each case.

$$2x-2y+z=\lambda x \quad , \quad 2x-3y+2z=\lambda y \ , \quad -x+2y=\lambda z$$

4. Verify the C-H Theorem of a matrix A= $\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and find A^{-1} also.

5. Diagonalise the matrix A= $\begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$ and obtained the Model matrix.